

ANDREA ASTON

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Summary

Final Year Electronics and Computer Engineering student (MUBAS, 2026) with hands-on experience in embedded systems, IoT, robotics, and full-stack web development. Delivered end-to-end technical projects from Raspberry Pi IoT systems to international competition robots and cloud-connected web platforms. I also do graphic designing, 3D modelling, PCB design.

Technical Skills

Languages:	Python, C++, JavaScript, HTML5, CSS3, SQL
Embedded / IoT:	Raspberry Pi, Arduino Kits, REV Robotics Kits,
Web & Cloud:	React, Supabase, REST APIs, Vercel, WordPress
CAD & Design:	SolidWorks, Fusion 360, KiCAD, Adobe PS/AI/Lr
Tools:	Git/GitHub, VS Code, Wokwi Simulator, LaTeX, Microsoft Office

Projects

Ac Graphic designing – I design posters, flyers, brochures etc. for students at Malawi University of Business and Applied Sciences (MUBAS) as well as Kamuzu University of Health Sciences as a side Hustle.

Full-Stack Portfolio Website · *React, Supabase, JavaScript, Vercel* · 2024–Present

- Designed and deployed a dynamic portfolio at andrea-aston.vercel.app with a headless CMS backend, CRUD admin dashboard, cloud image storage, and serverless contact form via EmailJS.

IoT Server Room Monitoring System · *Raspberry Pi 3, Python, Supabase, HTML/CSS/JS* · 2026

- Built a full IoT platform monitoring temperature, humidity, smoke, and grid power with automated fan/relay control and 5-second cloud telemetry to Supabase PostgreSQL.
- Developed a multi-user web dashboard with real-time refresh and authenticated access; implemented a 15-second anti-chatter power switching state machine for stable grid-to-battery failover.

First Global Challenge Robot — Team Malawi · *SolidWorks, REV Robotics, Python* · Panama 2025

- Designed and implemented FGC 2025 Team Malawi Robot

Submersible Pipe Inspection Robot · *SolidWorks, Resin 3D Printing* · 2024–2025

- Co-designed a waterproof submersible robot for borehole pipe inspection.

Work Experience

Robotics Technician Intern | Robotics Foundation Malawi *Jul 2024 – Present*

- Delivered 17-week STEM training program covering Arduino, REV Robotics, and competition autonomous programming; led FGC Team Malawi's full robot engineering effort through international competition.
- Established robotics clubs across two secondary schools in Blantyre through stakeholder engagement and live demonstrations; improved organizational technical literacy through staff embedded-systems workshops.

Web Development Intern | Corporate Tech *Feb – Jun 2024*

- Performed WordPress maintenance, theme customization, and content updates for client websites.

Education

BsC (Hons) Electronics and Computer Engineering *2022 – 2026 (Expected)*

Malawi University of Business and Applied Sciences (MUBAS), Blantyre

Relevant: Embedded Systems, Microprocessor Technology, Digital Electronics, Data Communication Networks, Software Engineering, OS, Signal Systems

MSCE — Loyola Jesuit Secondary School, Kasungu *2020*

Certifications

- **Google** — Foundations of Cybersecurity; Manage Security Risks (Coursera, 2023)
- **IBM** — Data Engineering, Cloud Computing, AI Chatbots (Coursera, 2023–24)
- **Meta** — Introduction to Databases (Cousera 2023)
- **HP LIFE** — AI for Business Professionals; AI for Beginners (2026)
- **Aspire Institute** — Leaders Program (2024)

REFEREES

- Mr Stanley Mbewe (Executive Director at Robotics Foundation Limited Malawi) | Phone: +265 991 54 86 57 | Email: Stanley@rflmw.com